

1 Final Form of the Vector Potential and the Emergence of Magnetic Moment

From the previous derivation, we arrived at the far-field expression of the vector potential of a small circular current loop as

$$\mathbf{A}(\mathbf{r}) = \hat{\phi} \frac{\mu_0 I a^2}{4r^2} \sin \theta \quad (1)$$

This is already a very important result. It shows that for a small loop observed far away:

- the vector potential has only an azimuthal component,
- it varies as $1/r^2$,
- and its angular dependence is $\sin \theta$.

Physical meaning of $\sin \theta$: The vector potential is maximum in the plane of the loop ($\theta = \pi/2$), and vanishes along the axis of the loop ($\theta = 0, \pi$). So the loop does not radiate its magnetic influence uniformly in all directions; its effect has a clear dipole-like angular nature.

Now rewrite (1) by introducing the area of the circular loop:

$$S = \pi a^2.$$

Then

$$Ia^2 = \frac{I(\pi a^2)}{\pi} = \frac{IS}{\pi}.$$

Substituting into (1), we obtain

$$\mathbf{A}(\mathbf{r}) = \hat{\phi} \frac{\mu_0}{4\pi} \frac{(IS) \sin \theta}{r^2}. \quad (2)$$

At this stage, the quantity IS naturally appears. This motivates the definition of the **magnetic moment** of the loop.

$$\mathbf{m} = IS \hat{\mathbf{n}} \quad (3)$$

where:

[leftmargin=5mm]

- I is the current in the loop,



- S is the area enclosed by the loop,
- $\hat{\mathbf{n}}$ is the unit normal to the loop, given by the right-hand rule.

For the present circular loop lying in the xy -plane,

$$\hat{\mathbf{n}} = \hat{\mathbf{z}}, \quad \mathbf{m} = I\pi a^2 \hat{\mathbf{z}}.$$

Hence, if $m = |\mathbf{m}| = I\pi a^2$, then (2) becomes

$$\mathbf{A}(\mathbf{r}) = \hat{\phi} \frac{\mu_0}{4\pi} \frac{m \sin \theta}{r^2} \quad (4)$$

Key insight about magnetic moment:

The magnetic moment is the single quantity that summarizes the magnetic action of a small current loop.

- Larger current $\Rightarrow$ stronger magnetic effect.
- Larger loop area $\Rightarrow$ stronger magnetic effect.
- Its direction tells the axis of the equivalent tiny magnet.

Thus, $\mathbf{m}$ tells both:

1. **how strong** the loop behaves as a magnet, and
2. **in which direction** that magnetic axis points.

So a small current loop, when viewed from far away, behaves like a **magnetic dipole**.

Right-hand-rule interpretation: Curl the fingers of your right hand in the direction of current around the loop. Then your thumb points in the direction of $\mathbf{m}$. So the magnetic moment is perpendicular to the plane of the loop, not along the wire.

Now let us express (4) in a compact vector form.

Since for this loop

$$\mathbf{m} = m \hat{\mathbf{z}},$$

we note that

$$\mathbf{m} \times \hat{\mathbf{r}} = m \sin \theta \hat{\phi}.$$

Therefore, (4) may be written as

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \frac{\mathbf{m} \times \hat{\mathbf{r}}}{r^2} \quad (5)$$



Using $\hat{\mathbf{r}} = \mathbf{r}/r$, this can also be written as

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \frac{\mathbf{m} \times \mathbf{r}}{r^3} \quad (6)$$

Final result: vector potential of a small current loop at a far-away point

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \frac{\mathbf{m} \times \mathbf{r}}{r^3}$$

This compact result shows that, at large distances, the entire loop is represented only through its magnetic moment $\mathbf{m}$. That is why magnetic moment is the most natural and powerful quantity to describe a small current loop.

What has been achieved so far: We started from the current loop, used symmetry to simplify the vector potential, applied the far-field approximation, and finally arrived at a dipole-like expression in terms of $\mathbf{m}$. The next natural step is to use

$$\mathbf{B} = \nabla \times \mathbf{A}$$

to obtain the magnetic field of the small loop in the far field.

1.1 Magnetic Field from the Derived Vector Potential

From the previous section, for a **small current loop observed far away**, the vector potential was obtained as

$$\mathbf{A}(\mathbf{r}) = \hat{\phi} \frac{\mu_0 m \sin \theta}{4\pi r^2} \quad (7)$$

where $m = |\mathbf{m}| = I\pi a^2$.

We now use

$$\mathbf{B} = \nabla \times \mathbf{A}. \quad (8)$$

Since $\mathbf{A}$ has only an azimuthal component,

$$A_r = 0, \quad A_\theta = 0, \quad A_\phi = \frac{\mu_0 m \sin \theta}{4\pi r^2},$$

it is convenient to evaluate the curl in spherical coordinates.

Curl in spherical coordinates

Using the determinant form,

$$\nabla \times \mathbf{A} = \frac{1}{r^2 \sin \theta} \begin{vmatrix} \hat{\mathbf{r}} & r\hat{\theta} & r \sin \theta \hat{\phi} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ 0 & 0 & r \sin \theta A_\phi \end{vmatrix}. \quad (9)$$

Substituting A_ϕ from (7),

$$r \sin \theta A_\phi = r \sin \theta \left(\frac{\mu_0 m \sin \theta}{4\pi r^2} \right) = \frac{\mu_0 m}{4\pi} \frac{\sin^2 \theta}{r}.$$

Therefore,

$$\mathbf{B} = \frac{1}{r^2 \sin \theta} \left\{ \hat{\mathbf{r}} \frac{\partial}{\partial \theta} (r \sin \theta A_\phi) - r\hat{\theta} \frac{\partial}{\partial r} (A_\phi r \sin \theta) \right\} \quad (10)$$

$$= \frac{1}{r^2 \sin \theta} \left\{ \hat{\mathbf{r}} \frac{\partial}{\partial \theta} \left(\frac{\mu_0 m}{4\pi} \frac{\sin^2 \theta}{r} \right) - r\hat{\theta} \frac{\partial}{\partial r} \left(\frac{\mu_0 m}{4\pi} \frac{\sin^2 \theta}{r} \right) \right\}. \quad (11)$$

Now evaluate the two derivatives one by one.

Radial component term:

$$\frac{\partial}{\partial \theta} \left(\frac{\mu_0 m}{4\pi} \frac{\sin^2 \theta}{r} \right) = \frac{\mu_0 m}{4\pi r} \frac{\partial}{\partial \theta} (\sin^2 \theta) \quad (12)$$

$$= \frac{\mu_0 m}{4\pi r} (2 \sin \theta \cos \theta). \quad (13)$$

Polar component term:

$$\frac{\partial}{\partial r} \left(\frac{\mu_0 m}{4\pi} \frac{\sin^2 \theta}{r} \right) = \frac{\mu_0 m \sin^2 \theta}{4\pi} \frac{\partial}{\partial r} \left(\frac{1}{r} \right) \quad (14)$$

$$= -\frac{\mu_0 m \sin^2 \theta}{4\pi r^2}. \quad (15)$$

Substituting back into (10),

$$\mathbf{B} = \frac{1}{r^2 \sin \theta} \left[\hat{\mathbf{r}} \frac{\mu_0 m}{4\pi r} (2 \sin \theta \cos \theta) + r\hat{\theta} \frac{\mu_0 m \sin^2 \theta}{4\pi r^2} \right] \quad (16)$$

$$= \frac{\mu_0 m}{4\pi} \left[\hat{\mathbf{r}} \frac{2 \cos \theta}{r^3} + \hat{\theta} \frac{\sin \theta}{r^3} \right]. \quad (17)$$



Hence, the magnetic field of a small current loop in the far field is

$$\mathbf{B} = \frac{\mu_0 m}{4\pi} \left[\hat{\mathbf{r}} \frac{2 \cos \theta}{r^3} + \hat{\theta} \frac{\sin \theta}{r^3} \right] \quad (18)$$

Magnetic field of a small current loop:

$$\mathbf{B} = \frac{\mu_0 m}{4\pi} \left[\hat{\mathbf{r}} \frac{2 \cos \theta}{r^3} + \hat{\theta} \frac{\sin \theta}{r^3} \right]$$

This is the far-field magnetic field of a small circular current loop.

Using the magnetic moment vector $\mathbf{m}$, the corresponding vector potential may also be written compactly as

$$\mathbf{A} = \frac{\mu_0}{4\pi} \frac{\mathbf{m} \times \mathbf{r}}{r^3} \quad (19)$$

Important note: Both expressions (18) and (19) are valid for a **small loop observed at a far-away point**. That is, they are obtained under the approximation

$$r \gg a.$$

Points to be noted

From the above two expressions, we observe:

- The vector potential varies as

$$\mathbf{A} \propto \frac{1}{r^2}.$$

- The magnetic field varies as

$$\mathbf{B} \propto \frac{1}{r^3}.$$

Comparison with the electrostatic dipole: This behavior is very similar to the electric dipole in electrostatics:

- electric potential due to a dipole varies as $1/r^2$,
- electric field due to a dipole varies as $1/r^3$.

So mathematically, the magnetic field of a small current loop has the same dipole-type spatial variation as the electric field of an electrostatic dipole.



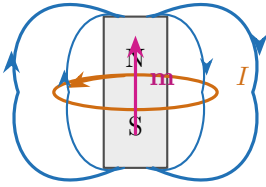
In other words,

the field due to a small current loop is dipole-like, just as the field due to an electric dipole is dipole-like.

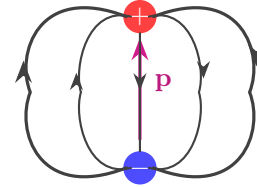
1.2 Comparison with the Electrostatic Dipole

The result derived for the small current loop strongly resembles the field of an **electric dipole**. To make this analogy clearer, let us place the two side by side.

Magnetic dipole



Electric dipole



B-field lines of a small loop

E-field lines of an electric dipole

Figure 1: Comparison of far-field dipole patterns. A small current loop behaves as a magnetic dipole with moment $\mathbf{m}$, while two equal and opposite charges form an electric dipole with moment $\mathbf{p}$. The field patterns are similar in shape, though their physical origin is different.

Main physical analogy:

- A **small current loop** behaves like a **magnetic dipole**.
- Two equal and opposite charges separated by a small distance form an **electric dipole**.
- In the far field, both have the same *angular structure* and the same *distance-decay law*, namely $1/r^3$ for the field.

1.3 Corresponding Equations for the Electric Dipole

For an electric dipole with dipole moment

$$\mathbf{p} = q \mathbf{d} \quad (20)$$



where $\mathbf{d}$ is directed from the negative charge to the positive charge, the electrostatic potential at a far-away point is

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{\mathbf{p} \cdot \mathbf{r}}{r^3} = \frac{1}{4\pi\epsilon_0} \frac{p \cos \theta}{r^2} \quad (21)$$

The corresponding electric field is

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0 r^3} [3(\mathbf{p} \cdot \hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{p}] \quad (22)$$

and, for $\mathbf{p} = p\hat{\mathbf{z}}$, this becomes in spherical coordinates

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \left[\hat{\mathbf{r}} \frac{2p \cos \theta}{r^3} + \hat{\theta} \frac{p \sin \theta}{r^3} \right]. \quad (23)$$

1.4 Why This Comparison is Important

This comparison helps us understand the role of magnetic moment much more clearly.

- In electrostatics, $\mathbf{p}$ summarizes the strength and direction of a pair of separated charges.
- In magnetostatics, $\mathbf{m}$ summarizes the strength and direction of a small circulating current loop.

Thus:

$$\text{Electric dipole moment } \mathbf{p} \text{ for charge separation} \quad \Longleftrightarrow \quad \text{Magnetic dipole moment } \mathbf{m} \text{ for current circulation}$$

A very useful way to remember:

- Static separated charges $\rightarrow$ electric dipole moment $\mathbf{p}$
- Circulating current in a small loop $\rightarrow$ magnetic dipole moment $\mathbf{m}$

Both create dipole-type fields in the far region.

1.5 Field Behavior Away from Sources

Notice that, away from the actual sources,

- for the electric dipole field, there is no charge in the observation region,



- for the magnetic dipole field of a small loop, there is no current in the observation region.

So in the source-free region, the fields behave smoothly and satisfy the appropriate source-free equations.

For example, for the small current loop far field,

$$\mathbf{B} = \frac{\mu_0 m}{4\pi} \left[\hat{\mathbf{r}} \frac{2 \cos \theta}{r^3} + \hat{\theta} \frac{\sin \theta}{r^3} \right],$$

which is valid only away from the current-carrying loop itself.

Important caution: The dipole expressions derived above are *far-field approximations*. They describe the field correctly when the observation point is far away compared to the size of the dipole. They must not be used blindly very close to the actual source distribution.

1.6 Curl of the Dipole Field Away from the Loop

From the far-field result already derived for a small current loop,

$$\mathbf{B} = \frac{\mu_0 m}{4\pi} \left[\frac{2 \cos \theta}{r^3} \hat{\mathbf{r}} + \frac{\sin \theta}{r^3} \hat{\theta} \right], \quad (24)$$

we now evaluate $\nabla \times \mathbf{B}$.

Since $\mathbf{B}$ has only r - and θ -components, and no ϕ -dependence, the curl in spherical coordinates reduces to only a $\hat{\phi}$ -component:

$$\nabla \times \mathbf{B} = \frac{1}{r} \left[\frac{\partial}{\partial r}(r B_\theta) - \frac{\partial B_r}{\partial \theta} \right] \hat{\phi}. \quad (25)$$

Using (24),

$$B_r = \frac{\mu_0 m}{4\pi} \frac{2 \cos \theta}{r^3}, \quad B_\theta = \frac{\mu_0 m}{4\pi} \frac{\sin \theta}{r^3}.$$

Therefore,

$$r B_\theta = \frac{\mu_0 m}{4\pi} \frac{\sin \theta}{r^2},$$

and hence

$$\frac{\partial}{\partial r}(r B_\theta) = \frac{\mu_0 m}{4\pi} \sin \theta \frac{\partial}{\partial r} \left(\frac{1}{r^2} \right) = -\frac{\mu_0 m}{4\pi} \frac{2 \sin \theta}{r^3}. \quad (26)$$

Also,

$$\frac{\partial B_r}{\partial \theta} = \frac{\mu_0 m}{4\pi} \frac{\partial}{\partial \theta} \left(\frac{2 \cos \theta}{r^3} \right) = -\frac{\mu_0 m}{4\pi} \frac{2 \sin \theta}{r^3}. \quad (27)$$



Substituting (26) and (27) into (25),

$$\nabla \times \mathbf{B} = \frac{1}{r} \left[-\frac{\mu_0 m}{4\pi} \frac{2 \sin \theta}{r^3} - \left(-\frac{\mu_0 m}{4\pi} \frac{2 \sin \theta}{r^3} \right) \right] \hat{\phi} = 0.$$

Thus,

$$\boxed{\nabla \times \mathbf{B} = 0 \quad \text{for } r \neq 0} \quad (28)$$

Important concept: For a *small current loop*, the far-field magnetic field satisfies

$$\boxed{\nabla \times \mathbf{B} = 0}$$

everywhere away from the actual current-carrying region.

So the magnetic field may be nonzero far away, but its *curl* is zero there because no current passes through that observation point.

Do not confuse these two statements:

- $\mathbf{B} \neq 0$ far away from the loop,
- but $\nabla \times \mathbf{B} = 0$ there.

A field can be present at a point due to distant sources, while its curl at that point is still zero because there is no *local current density* there.

1.7 Why This Result is Important

The above result is not just a property of one small loop. It reveals a much deeper idea in magnetostatics.

Key physical message: A magnetic field at a point can be produced by current elements *everywhere*, but the quantity

$$\nabla \times \mathbf{B}$$

responds only to the *current actually passing through the local neighborhood of that point*.

This is the magnetic analogue of what we already know in electrostatics:

- the electric field $\mathbf{E}$ may be caused by charges far away,
- but the divergence $\nabla \cdot \mathbf{E}$ sees only the *local charge density*.

Similarly, in magnetostatics:



- the magnetic field $\mathbf{B}$ is due to current everywhere,
- but the curl $\nabla \times \mathbf{B}$ sees only the *local current density*.

1.8 From a Tiny Loop to a General Current Distribution

A small current loop is the simplest magnetic dipole element. That is why tiny loops are the natural building blocks for understanding magnetic dipole behavior and, more broadly, magnetic materials.

Now think of a general steady current distribution $\mathbf{J}$.

At a given observation point:

- current elements far away contribute to $\mathbf{B}$, but around that point they behave like source-free contributions, so their local curl is zero;
- only current passing through the immediate neighborhood of that point contributes to $\nabla \times \mathbf{B}$.

Hence, for a general steady current distribution,

$$\boxed{\nabla \times \mathbf{B} = \mu_0 \mathbf{J}} \quad (29)$$

which is Ampère's law in differential form.

Very important interpretation of Ampère's law:

$$\boxed{\mathbf{B} \text{ is due to currents everywhere,} \quad \nabla \times \mathbf{B} \text{ sees only local current.}}$$

1.9 Analogy with Electrostatics

Let us now place the magnetic and electrostatic results side by side.

In electrostatics:

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \iiint \rho(\mathbf{r}') \frac{\mathbf{R}}{R^3} dv', \quad (30)$$

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \iiint \frac{\rho(\mathbf{r}')}{R} dv', \quad (31)$$

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}. \quad (32)$$



In magnetostatics:

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \iiint \frac{\mathbf{J}(\mathbf{r}') \times \mathbf{R}}{R^3} dv', \quad (33)$$

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \iiint \frac{\mathbf{J}(\mathbf{r}')}{R} dv', \quad (34)$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}. \quad (35)$$

Here,

$$\mathbf{R} = \mathbf{r} - \mathbf{r}', \quad R = |\mathbf{r} - \mathbf{r}'|.$$

Beautiful parallel structure:

$$V \longleftrightarrow \mathbf{A}, \quad \mathbf{E} \longleftrightarrow \mathbf{B}, \quad \nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0} \longleftrightarrow \nabla \times \mathbf{B} = \mu_0 \mathbf{J}$$

So electrostatics and magnetostatics have a very similar mathematical structure, but with different kinds of sources.

1.10 Integral Forms and Their Meanings

From (29), using Stokes' theorem,

$$\oint_C \mathbf{B} \cdot d\boldsymbol{\ell} = \iint_S (\nabla \times \mathbf{B}) \cdot d\mathbf{s} \quad (36)$$

$$= \mu_0 \iint_S \mathbf{J} \cdot d\mathbf{s} \quad (37)$$

$$= \mu_0 I_{\text{encl}}. \quad (38)$$

Thus,

$$\oint_C \mathbf{B} \cdot d\boldsymbol{\ell} = \mu_0 I_{\text{encl}} \quad (39)$$

which is Ampère's circuital law.

Similarly, from Gauss's law in electrostatics,

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0},$$

using the divergence theorem,

$$\oint_S \mathbf{E} \cdot d\mathbf{s} = \iiint_V (\nabla \cdot \mathbf{E}) dv \quad (40)$$

$$= \iiint_V \frac{\rho}{\epsilon_0} dv \quad (41)$$

$$= \frac{q_{\text{encl}}}{\epsilon_0}. \quad (42)$$

Hence,

$$\oint_S \mathbf{E} \cdot d\mathbf{s} = \frac{q_{\text{encl}}}{\epsilon_0} \quad (43)$$

Final summary:

- Tiny current loops are the basic dipole elements of magnetostatics.
- Away from the loop, $\nabla \times \mathbf{B} = 0$.
- For a general current distribution,

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}.$$

- Magnetic field is due to current everywhere, but its curl detects only *local* current density.
- This is closely analogous to electrostatics, where $\mathbf{E}$ is due to charge everywhere, but $\nabla \cdot \mathbf{E}$ detects only *local* charge density.

2 From a Small Current Loop to an Infinitely Long Solenoid

A solenoid can be viewed as a dense collection of many current-carrying loops. Motivated by the discussion of the *small current loop*, let us now consider an **infinitely long cylinder of arbitrary cross-section**, uniformly wound by a current-carrying coil with n turns per unit length, as shown in Fig. 2. We first identify the *allowed form* of the vector potential before actually solving for the magnetic field.

To connect this problem with the previous vector-potential formulation, recall

$$\mathbf{B} = \nabla \times \mathbf{A}, \quad (44)$$

and

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int_V \frac{\mathbf{J}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} dV'. \quad (45)$$

Now we use symmetry very carefully.



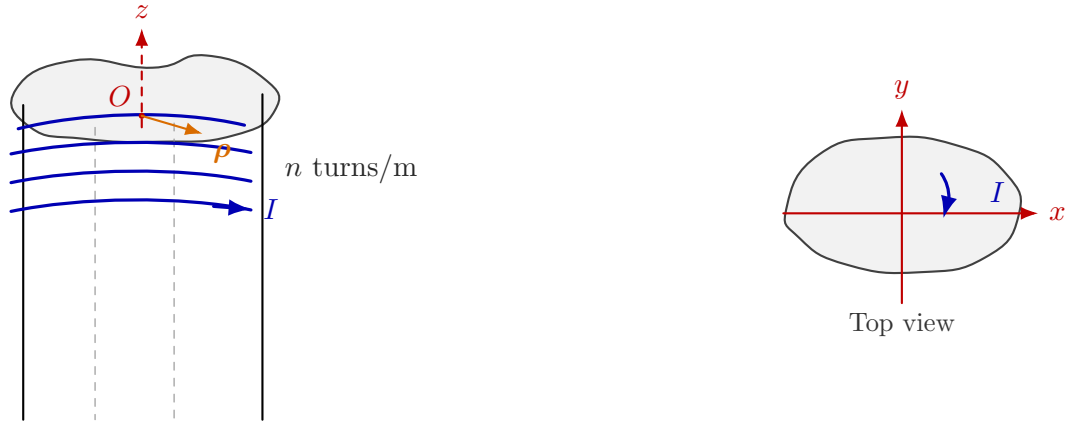


Figure 2: An infinitely long cylinder of arbitrary cross-section, uniformly wound by a current-carrying coil. The right sketch shows the transverse cross-section. Since the winding goes *around* the cylinder, the source current has no axial ($\hat{z}$) component.

Observation 1: translational symmetry along the axis. Since the cylinder is assumed to be **infinitely long** and uniformly wound, the geometry does not change along the axial direction. Therefore,

$$\frac{\partial}{\partial z}(\cdot) = 0, \quad (46)$$

so the electromagnetic quantities do not depend on z .

Observation 2: the winding current is not along $\hat{z}$. The current in the coil wraps *around* the cylinder. Hence the source current has **no axial component**:

$$J_z = 0. \quad (47)$$

So the current density is confined to the transverse plane.

Therefore, before solving for the field, the vector potential must also be taken in a form consistent with this symmetry:

$$\mathbf{A}(\mathbf{r}) = A_x(x, y) \hat{\mathbf{x}} + A_y(x, y) \hat{\mathbf{y}}, \quad A_z = 0. \quad (48)$$

If we describe the transverse plane schematically in polar form, the same statement may be written as

$$\mathbf{A}(\mathbf{r}) = A_\rho(\rho, \phi) \hat{\boldsymbol{\rho}} + A_\phi(\rho, \phi) \hat{\boldsymbol{\phi}}, \quad A_z = 0. \quad (49)$$

Thus, at this stage, the key point is:

For an **infinitely long, uniformly wound solenoid**, there is **no variation along z** and the source current has **no z -component**. Hence we begin with a **purely transverse vector potential $\mathbf{A}$** , and only then proceed to determine the magnetic field.



Curl of $\mathbf{A}$ in cylindrical coordinates. Since the geometry is naturally cylindrical, let us now evaluate $\mathbf{B} = \nabla \times \mathbf{A}$ in cylindrical coordinates. From the previous discussion, we have

$$\mathbf{A}(\mathbf{r}) = A_\rho(\rho, \phi) \hat{\rho} + A_\phi(\rho, \phi) \hat{\phi}, \quad A_z = 0,$$

and there is no dependence on z .

Therefore,

$$\mathbf{B} = \nabla \times \mathbf{A} = \frac{1}{\rho} \begin{vmatrix} \hat{\rho} & \rho \hat{\phi} & \hat{z} \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_\rho(\rho, \phi) & \rho A_\phi(\rho, \phi) & 0 \end{vmatrix}. \quad (50)$$

Now observe the following:

- $A_z = 0$,
- $\frac{\partial}{\partial z}(\cdot) = 0$ because the solenoid is infinitely long,
- hence the $\hat{\rho}$ and $\hat{\phi}$ components of $\nabla \times \mathbf{A}$ vanish.

So only the z -component survives:

$$\mathbf{B}(\mathbf{r}) = \frac{1}{\rho} \left[\frac{\partial}{\partial \rho}(\rho A_\phi(\rho, \phi)) - \frac{\partial A_\rho(\rho, \phi)}{\partial \phi} \right] \hat{z}. \quad (51)$$

Hence, the magnetic field must be of the form

$$\mathbf{B}(\rho, \phi) = \hat{z} B_z(\rho, \phi). \quad (52)$$

Thus, even before applying Ampère's law, symmetry and the curl operation together show that the magnetic field can only be directed along the axis of the solenoid.

2.1 Magnetic Field Due to a Symmetric Solenoid

We now consider a **symmetric solenoid**, namely an infinitely long cylindrical solenoid of **circular cross-section** and radius a , uniformly wound with n turns per unit length, each carrying current I . This is a **special case** of the more general arbitrary-cross-section problem discussed above, but the circular symmetry now allows us to simplify the form of the magnetic field even further.

Because this is a **circularly symmetric** cylinder, the field cannot depend on the azimuthal angle ϕ . From the previous result, we already know that the magnetic field can only be along $\hat{z}$. Therefore, the symmetry is now stronger, and we conclude that

$$\mathbf{B}(\mathbf{r}) = \hat{z} B_z(\rho). \quad (53)$$



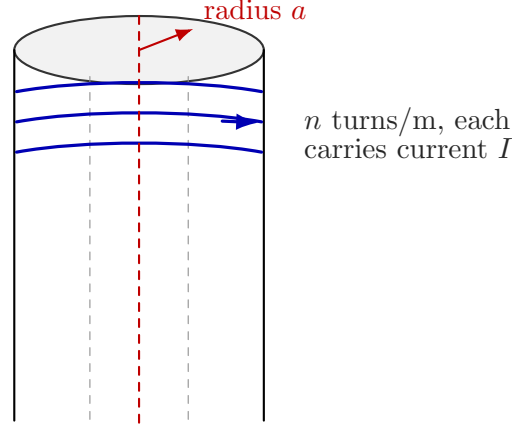


Figure 3: A symmetric solenoid: an infinitely long circular cylinder of radius a , uniformly wound with n turns per unit length, each carrying current I .

Thus, for the symmetric solenoid, the magnetic field depends only on the radial distance ρ from the axis. Since we now have enough symmetry, we can proceed directly to apply **Ampère's law**.

2.1.1 Applying Ampère's Law to the Symmetric Solenoid

From the previous section, for the symmetric solenoid we already established that

$$\mathbf{B}(\mathbf{r}) = \hat{\mathbf{z}} B_z(\rho). \quad (54)$$

Since we now have enough symmetry, Ampère's law becomes the natural tool:

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}, \quad \oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{encl}}. \quad (55)$$

To exploit this symmetry, choose a rectangular Amperian loop in the ρ - z plane, of *unit height*, extending from the axis ($\rho = 0$) to an arbitrary radial location $\rho = r$, as shown in Fig. 4.

Because $\mathbf{B}$ is purely along $\hat{\mathbf{z}}$, the two horizontal sides of the rectangular loop give no contribution:

$$\mathbf{B} \cdot d\mathbf{l} = 0 \quad \text{on the radial segments.}$$

Thus only the two vertical sides contribute. For a loop of unit height,

$$\oint \mathbf{B} \cdot d\mathbf{l} = B_z(0) + 0 - B_z(r) + 0. \quad (56)$$

Next, the enclosed current depends on whether the loop cuts through the winding sheet at $\rho = a$:

$$I_{\text{encl}} = \begin{cases} 0, & r < a, \\ nI, & r > a. \end{cases} \quad (57)$$

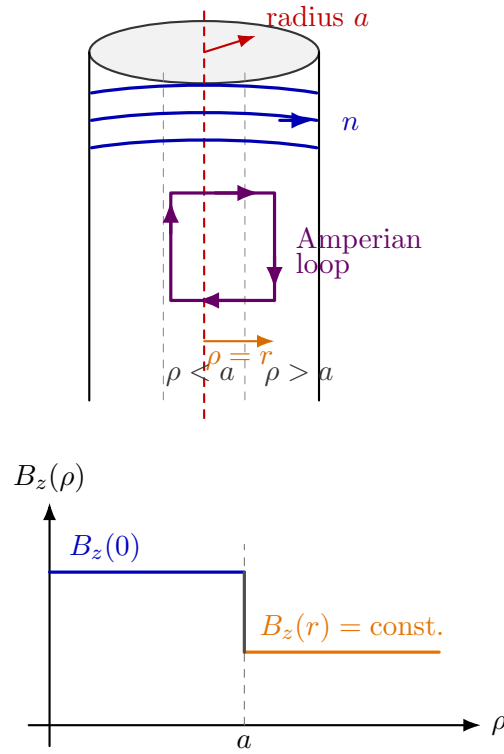


Figure 4: Rectangular Amperian loop of unit height used for the symmetric solenoid. The same construction already shows that B_z is constant for $\rho < a$ and also constant for $\rho > a$, though these two constants need not be the same.

Therefore, from Ampère's law,

$$B_z(0) - B_z(r) = \mu_0 I_{\text{encl}} = \begin{cases} 0, & r < a, \\ \mu_0 n I, & r > a. \end{cases} \quad (58)$$

Equivalently,

$$B_z(r) = B_z(0), \quad r < a, \quad (59)$$

and

$$B_z(r) = B_z(0) - \mu_0 n I, \quad r > a. \quad (60)$$

So, up to this stage, Ampère's law tells us that the magnetic field is **constant inside** the solenoid and also **constant outside**, with a jump of magnitude $\mu_0 n I$ across the current sheet at $\rho = a$.

Determining the outside field. From Eq. (60), we found that for $\rho > a$, the magnetic field is a *constant*:

$$B_z(\rho) = B_z(0) - \mu_0 n I, \quad \rho > a.$$

But physically, very far away from the solenoid, the magnetic field must vanish. This is consistent with the fact that the field due to a localized current loop decreases rapidly with distance ($|\mathbf{B}| \propto 1/r^3$ in the far field). Therefore, for an ideal infinitely long solenoid, the only acceptable constant value outside is

$$\boxed{\mathbf{B}(\rho) = 0, \quad \rho > a.} \quad (61)$$

Substituting this into Eq. (60), we get

$$0 = B_z(0) - \mu_0 n I, \quad (62)$$

so that

$$\boxed{B_z(0) = \mu_0 n I.} \quad (63)$$

Now, from Eq. (59), the field inside the solenoid is already known to be constant. Hence,

$$\boxed{\mathbf{B}(\rho) = \mu_0 n I \hat{\mathbf{z}}, \quad \rho < a.} \quad (64)$$

Therefore, for an **ideal infinitely long solenoid**,

$$\mathbf{B}(\rho) = \begin{cases} \mu_0 n I \hat{\mathbf{z}}, & \rho < a, \\ 0, & \rho > a. \end{cases}$$

That is, the magnetic field is **uniform inside** and **zero outside**.

2.2 Example 2: Torus

A torus (or toroid) may be viewed as a **solenoid bent into a circular shape**. We now study the magnetic field due to such a toroidal winding.

Symmetry of the current density. Because the torus is rotationally symmetric about the z -axis, the current distribution is **independent of the azimuthal coordinate ϕ** . Hence,

$$\frac{\partial}{\partial \phi}(\cdot) = 0, \quad \mathbf{J} = \mathbf{J}(\rho, z). \quad (65)$$

Also, for each turn, the wire wraps around the *cross-section* of the torus, i.e. the current flows only in the ρ - and z -directions. Therefore,

$$\boxed{J_\phi = 0.} \quad (66)$$

So, in cylindrical coordinates,

$$\boxed{\mathbf{J}(\rho, z) = J_\rho(\rho, z) \hat{\boldsymbol{\rho}} + J_z(\rho, z) \hat{\mathbf{z}}.} \quad (67)$$



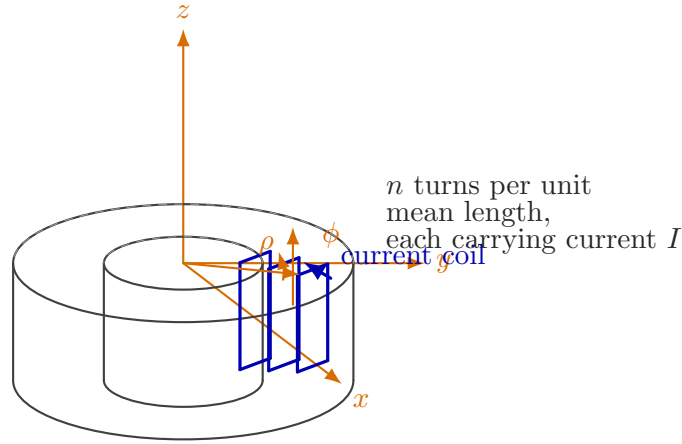


Figure 5: A torus may be viewed as a solenoid bent into a circular shape. We use cylindrical coordinates (ρ, ϕ, z) about the symmetry axis z .

First step, as in the previous problem. Before solving for the magnetic field, we must first determine *which components* of the vector potential $\mathbf{A}$ are allowed.

However, unlike the previous solenoid problem, this is **not immediately obvious**. One might be tempted to say:

$$J_\phi = 0 \implies A_\phi = 0.$$

But this conclusion does *not* follow directly here.

Reason. Recall the general expression

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int_V \frac{\mathbf{J}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} dV'. \quad (68)$$

Now $\mathbf{J}(\mathbf{r}')$ is specified at the *source point* $\mathbf{r}'$, in terms of the local basis vectors $\hat{\rho}'$, $\hat{\phi}'$, and $\hat{\mathbf{z}}$. These basis directions are not the same as the basis vectors at the observation point $\mathbf{r}$.

Therefore, even though $J_\phi = 0$, we cannot immediately decide which component(s) of $\mathbf{A}$ vanish. That must be established separately from symmetry.

Why $A_\phi = 0$ is *not* obvious a priori. For the torus, we already know that the current density is independent of ϕ , and that

$$\mathbf{J}(\mathbf{r}') = J_\rho(\rho', z') \hat{\rho}' + J_z(\rho', z') \hat{\mathbf{z}}, \quad J_\phi = 0.$$

However, from this alone, we *cannot* immediately conclude that $A_\phi = 0$. The reason is that the vector potential is obtained from

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int_V \frac{\mathbf{J}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} dV', \quad (69)$$



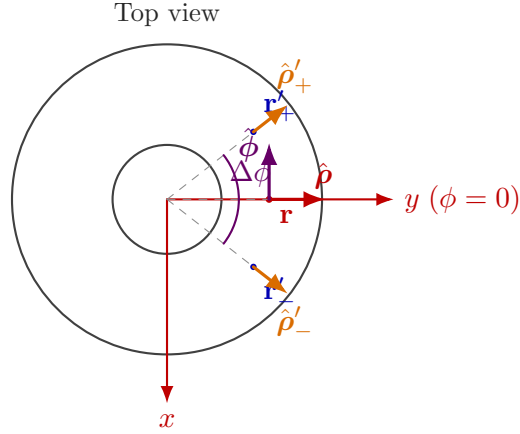


Figure 6: Top view of the torus. Even though $J_\phi = 0$, the source direction $\hat{\rho}'$ is not the same as the observation-point direction $\hat{\rho}$. Hence a single source contribution need not be purely along $\hat{\rho}$ and $\hat{z}$.

and the source basis vector $\hat{\rho}'$ at $\mathbf{r}'$ is generally *not parallel* to the basis vector $\hat{\rho}$ at the observation point $\mathbf{r}$.

Consider the observation point $\mathbf{r}$ on the plane $\phi = 0$. For a *single* source point $\mathbf{r}'$, the radial source direction can be written in the basis at the observation point as

$$\hat{\rho}' = \cos(\phi' - \phi) \hat{\rho} + \sin(\phi' - \phi) \hat{\phi}. \quad (70)$$

Therefore, the contribution due to $J_\rho(\mathbf{r}')\hat{\rho}'$ generally contains a $\hat{\phi}$ -component. So the statement

$$J_\phi = 0 \implies A_\phi = 0$$

is **not** obvious from a single source contribution.

Now use symmetry of the full torus. Take two source points $\mathbf{r}'_+$ and $\mathbf{r}'_-$ that are symmetric about the plane $\phi = 0$. Because the torus is rotationally symmetric,

- the two source points are at equal distance from $\mathbf{r}$,
- their current magnitudes are the same,
- their $\hat{\phi}$ -contributions are equal in magnitude and opposite in sign.

Indeed, for the symmetric pair,

$$\hat{\rho}'_+ = \cos(\Delta\phi) \hat{\rho} + \sin(\Delta\phi) \hat{\phi}, \quad (71)$$

$$\hat{\rho}'_- = \cos(\Delta\phi) \hat{\rho} - \sin(\Delta\phi) \hat{\phi}. \quad (72)$$



Hence, when the two contributions are added, the $\hat{\phi}$ -parts cancel, while the $\hat{\rho}$ -parts add.

Also, the $J_z \hat{\mathbf{z}}$ contribution is always along $\hat{\mathbf{z}}$, since $\hat{\mathbf{z}}$ is the same everywhere.

Therefore, after integrating over the entire torus, the vector potential can only have $\hat{\rho}$ - and $\hat{\mathbf{z}}$ -components:

$$\boxed{\mathbf{A}(\mathbf{r}) = A_\rho(\rho, z) \hat{\rho} + A_z(\rho, z) \hat{\mathbf{z}}, \quad A_\phi = 0.} \quad (73)$$

Thus, $A_\phi = 0$ is **not** obtained from the local statement $J_\phi = 0$ alone. It follows only after using the **pairwise symmetry** of the complete toroidal current distribution.

Hence, the allowed form of $\mathbf{A}$ for the torus is From the symmetry argument established above, the vector potential can only have $\hat{\rho}$ - and $\hat{\mathbf{z}}$ -components, and because of axial symmetry it is independent of ϕ . Therefore,

$$\boxed{\mathbf{A}(\mathbf{r}) = A_\rho(\rho, z) \hat{\rho} + A_z(\rho, z) \hat{\mathbf{z}}.} \quad (74)$$

That is, the contribution comes only from the ρ - and z -components.

Now compute $\mathbf{B} = \nabla \times \mathbf{A}$. Using cylindrical coordinates,

$$\mathbf{B}(\mathbf{r}) = \nabla \times \mathbf{A} = \frac{1}{\rho} \begin{vmatrix} \hat{\rho} & \rho \hat{\phi} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_\rho(\rho, z) & 0 & A_z(\rho, z) \end{vmatrix}. \quad (75)$$

Since

$$A_\phi = 0, \quad \frac{\partial}{\partial \phi}(\cdot) = 0,$$

the $\hat{\rho}$ - and $\hat{\mathbf{z}}$ -components vanish, and only the $\hat{\phi}$ -component survives. Hence,

$$\mathbf{B}(\mathbf{r}) = \left[\frac{\partial A_\rho(\rho, z)}{\partial z} - \frac{\partial A_z(\rho, z)}{\partial \rho} \right] \hat{\phi}. \quad (76)$$

Therefore,

$$\boxed{\mathbf{B}(\mathbf{r}) = B_\phi(\rho, z) \hat{\phi}.} \quad (77)$$

Thus, for the symmetric torus, the magnetic field can only be in the $\hat{\phi}$ -direction.

Remark. This conclusion relies on the **full axial symmetry** of the toroidal current distribution. If the winding pattern were not symmetric in ϕ (for example, if one part of the torus had a different current arrangement from another part), then this simplification would no longer hold.



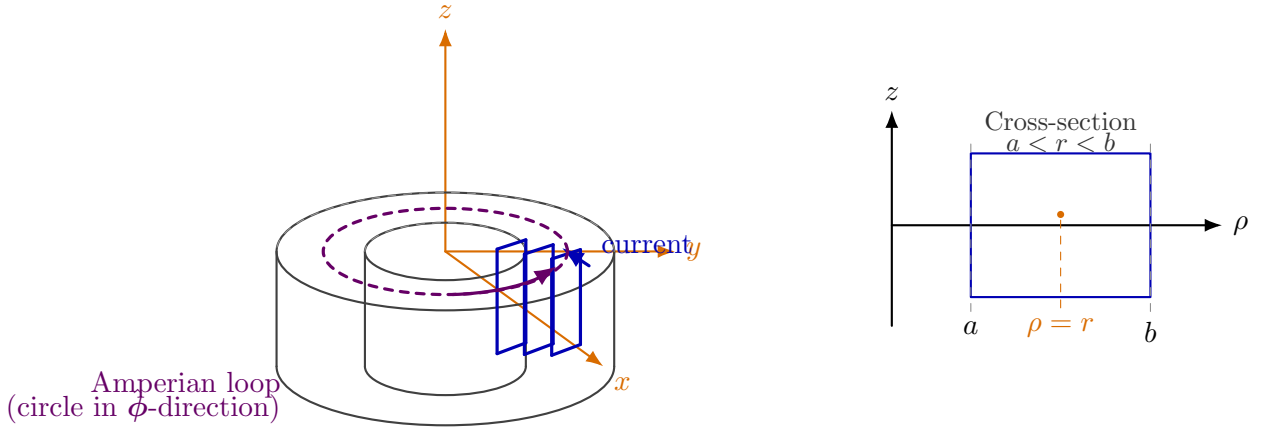


Figure 7: For the torus, the natural Amperian path is a circle centered on the symmetry axis. The same argument works for any point in the cross-section, provided the cross-section is independent of ϕ .

2.2.1 Applying Ampère's Law to the Torus

From the previous discussion, the magnetic field for the torus can only be along the azimuthal direction:

$$\mathbf{B}(\mathbf{r}) = B_\phi(\rho, z) \hat{\phi}. \quad (78)$$

Since the geometry is symmetric with respect to ϕ , the natural Amperian loop is a **circle in the $\hat{\phi}$ -direction**, centered on the z -axis and passing through the observation point, as shown in Fig. 7.

Field inside the toroidal winding. Consider a circular Amperian loop of radius r such that it lies within the winding region, that is,

$$a < r < b,$$

where a and b are the inner and outer radial limits of the torus.

Along this loop,

$$d\mathbf{l} = r d\phi \hat{\phi},$$

and since $\mathbf{B}$ is also along $\hat{\phi}$ and is independent of ϕ ,

$$\oint \mathbf{B} \cdot d\mathbf{l} = \int_0^{2\pi} B_\phi(r) r d\phi = 2\pi r B_\phi(r). \quad (79)$$

If the toroid has N total turns, each carrying current I , then the enclosed current is

$$I_{\text{encl}} = NI. \quad (80)$$

Therefore, from Ampère's law,

$$2\pi r B_\phi(r) = \mu_0 NI. \quad (81)$$



Hence,

$$\mathbf{B}(r) = \frac{\mu_0 N I}{2\pi r} \hat{\phi}, \quad a < r < b. \quad (82)$$

If the turn density along the mean circular path of radius c is n turns per unit length, then

$$N = (2\pi c)n,$$

and Eq. (82) may also be written as

$$\mathbf{B}(r) = \frac{\mu_0 n c I}{r} \hat{\phi}, \quad a < r < b. \quad (83)$$

Field outside the torus. Now use the *same* circular Amperian loop, but choose it outside the winding region.

Case 1: central hole of the torus ($r < a$). The surface bounded by the loop does not cut any turn of the winding. Hence,

$$I_{\text{encl}} = 0.$$

So,

$$2\pi r B_\phi(r) = 0 \implies \boxed{\mathbf{B}(r) = 0, \quad r < a.} \quad (84)$$

Case 2: outside the torus ($r > b$). Again consider the circular loop of radius r . The bounded surface is now pierced by the windings in such a way that the positive and negative crossings cancel in pairs. Therefore, the *net* enclosed current is still zero:

$$I_{\text{encl}} = 0.$$

Thus,

$$2\pi r B_\phi(r) = 0 \implies \boxed{\mathbf{B}(r) = 0, \quad r > b.} \quad (85)$$

Therefore, for an ideal toroidal winding,

$$\mathbf{B}(r) = \begin{cases} 0, & r < a, \\ \frac{\mu_0 N I}{2\pi r} \hat{\phi}, & a < r < b, \\ 0, & r > b. \end{cases}$$

So the magnetic field is confined to the interior of the toroidal winding.

Important remark. The cross-section need not be rectangular. As long as the toroid is symmetric with respect to the ϕ -direction, the same Amperian-loop argument applies, and the magnetic field inside the winding region remains

$$\mathbf{B}(r) = \frac{\mu_0 N I}{2\pi r} \hat{\phi}.$$